TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Identification of Fibers from a Given Sample (Single or Blended)

Burning, microscopy, solubility and selective dissolution — the systematic scheme that names the fibres in an unknown yarn or fabric, qualitatively and quantitatively.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 11 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to identify the fibre or fibres present in a given sample by a systematic scheme — preliminary examination and burning test, microscopic examination (longitudinal and cross-section), and solubility behaviour — and, for a two-component blend, to determine the blend proportions quantitatively by selective dissolution in the principle of ISO 1833 (quantitative chemical analysis of fibre mixtures).

1.1 Why identification must be systematic

Fibre content decides how a fabric is dyed, finished, laundered, priced and labelled; labelling is also a legal requirement in most markets, and a mis-declared blend is a trading offence as well as a technical error. Yet fibres cannot reliably be named by handle or appearance — mercerised cotton looks silk-like and textured polyester imitates wool's bulk. No single test names every fibre: burning groups them, microscopy separates many within a group, and solubility decides the cases left open. The scheme is therefore applied in order, cheapest first, and a conclusion is drawn only where the tests agree. Dyes and resins can mask burning odour and slow dissolution, so a preliminary scour is standard before quantitative work.

1.2 The burning test

A small tuft is observed in three stages — approaching the flame, in the flame, and after removal — with its odour and residue noted. The test reads the fibre's chemistry: cellulose burns like paper, proteins self-extinguish and smell of burnt hair, thermoplastics melt before they burn.

FIBRE	APPROACHING / IN FLAME	ON REMOVAL · ODOUR	RESIDUE
Cotton	Ignites at once; burns quickly, no shrink-away	Continues burning; burning-paper smell	Fine, soft grey ash; no bead

Viscose	Like cotton — ignites at once, burns quickly	Continues burning; burning-paper smell	Very little soft ash
Wool	Curls away; burns slowly, sizzling	Self-extinguishing; strong burnt-hair smell	Brittle black mass, crushes easily
Silk	Curls away slightly; burns slowly	Usually self-extinguishing; burnt-hair smell, milder than wool	Small shiny brittle black bead, crushable
Polyester	Shrinks and melts away; burns slowly, sooty flame	Usually self-extinguishing; sweet chemical smell	Hard dark bead, cannot be crushed
Nylon (polyamide)	Shrinks and melts away; burns very slowly, may drip	Usually self-extinguishing; celery-like chemical smell	Hard grey-brown bead, cannot be crushed
Acrylic	Melts, shrinks; burns rapidly, sputtering	Continues burning; acrid chemical smell	Hard, irregular black bead
Acetate (solubility contrast)	Melts and burns	Continues; vinegar smell	Hard dark bead

The table also teaches the test's limits. Cotton and viscose are indistinguishable by burning — both are cellulose — and must be separated by microscopy. In a blend, the major component's burning can mask the minor one, and a flame-retardant finish can make cellulose behave as self-extinguishing. Burning therefore assigns a family, never a final composition.

1.3 Microscopy

Under the microscope each fibre carries a structural signature, seen longitudinally and in cross-section. **Cotton** is a flat, twisted ribbon with convolutions and a kidney-shaped cross-section with a central lumen. **Wool** shows overlapping surface scales and a round-to-oval cross-section, sometimes with a medulla in coarse fibres. **Silk** is a smooth, featureless rod, triangular in cross-section. **Viscose** shows longitudinal striations and a serrated, lobed cross-section — the skin of a regenerated filament — which separates it from cotton at once. **Manufactured fibres** such as polyester and nylon are smooth, uniform rods with round cross-sections unless deliberately made trilobal or hollow; their uniformity identifies them as man-made, and burning plus solubility separate polyester from nylon. Draw each view, do not merely describe it: the drawing is the record that the features were actually seen.

1.4 Solubility and confirmatory tests

Solubility applies chemical distinctions that appearance cannot. The well-established classroom facts — stated carefully, because concentration, temperature and time are part of each test — are these. **Acetone** dissolves acetate at room temperature and leaves the common apparel fibres untouched, so an acetone treatment that removes a component identifies acetate. **Hot 5% sodium hydroxide** (boiling dilute alkali) dissolves wool — proteins are destroyed by hot alkali — and is the classic route for separating wool from cellulose fibres, which resist it; silk, also a protein, is likewise attacked, so alkali behaviour groups the proteins rather than separating wool from silk. **Concentrated sulfuric acid** dissolves cellulose and also attacks polyamide, and is used in standardised blend methods to dissolve the cellulose of a cotton/polyester blend, leaving polyester as residue — the chemistry behind the worked example below. A **staining test** confirms in outline: a formulated identification stain colours the fibre families differently, giving a rapid cross-check before the conclusion is signed off. Solvents are handled in the fume cupboard, exactly as the safety sheet directs.

1.5 Quantitative analysis by selective dissolution (ISO 1833 principle)

Once the components are known, ISO 1833 determines how much of each is present: a reagent dissolves one component completely while leaving the other essentially intact; the residue is washed, dried and weighed, and the composition follows by mass. Because no reagent is perfectly selective, accredited methods apply small published correction factors for mass lost by the "insoluble" component; a teaching determination states whether corrections were applied.

$$\text{Component dissolved (\%)} = (\text{initial dry mass} - \text{dry residue mass}) / \text{initial dry mass} \times 100$$

$$\text{Residue component (\%)} = \text{dry residue mass} / \text{initial dry mass} \times 100$$

The analysis is done on dry mass — specimens are dried to constant mass before and after treatment, since retained moisture would otherwise be reported as fibre. Duplicates that disagree beyond the method's tolerance are repeated, not averaged into respectability.

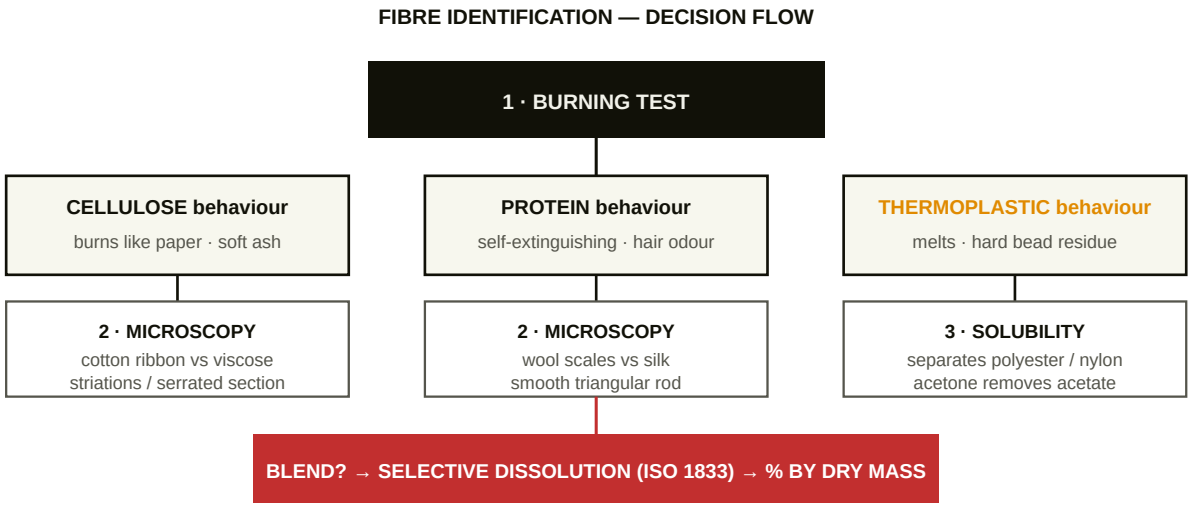


Figure 1 — The identification scheme as a flow. Burning assigns the family; microscopy separates the natural fibres within each family; solubility decides the manufactured fibres and confirms the rest; a known blend then goes to selective dissolution for its percentages.

2. Apparatus and procedure

- **Apparatus:** forceps and a small spirit flame for the burning test; compound microscope with slides, cover slips and mounting medium, and a microtome or razor method for cross-sections; reagents as directed — acetone, 5% sodium hydroxide, concentrated sulfuric acid (fume cupboard, safety sheet) — beakers, filter crucibles; drying oven and balance (0.001 g); identification stain.
- **Sampling:** tease a representative tuft from several places in the yarn or fabric; for fabric, separate warp and weft if they may differ. Remove surface finish from the quantitative sub-sample by the preliminary scour specified, then dry to constant mass.
- **Burning test:** hold a small tuft in forceps: approach the flame edge (note shrinking or curling away), enter the flame (note burning character), then withdraw it (note continued burning or self-extinction, and odour — wafted, never inhaled directly). Crush the cooled residue: soft ash, crushable brittle mass, or uncrushable hard bead. Record all observations before consulting the table.
- **Microscopy:** mount fibres in liquid paraffin or water, examine longitudinally and draw the view; prepare or examine a cross-section and draw it. Compare feature by feature — convolutions, scales, striations, section shape — against the theory descriptions.
- **Solubility:** place small tufts in each selected reagent under the stated conditions (concentration, temperature, time) and record dissolves / swells / unaffected. Run a known reference fibre beside an uncertain result whenever one is available.
- **Quantitative determination (cotton/polyester):** weigh the dried specimen, treat with the sulfuric-acid reagent under the standardised conditions so the cellulose dissolves, filter, wash the residue

until washings are neutral, dry to constant mass and weigh. Calculate both percentages by dry mass; run a duplicate; state that standard correction factors are applied in accredited work.

- **Conclusion:** write the identification as a chain of evidence — family by burning, species by microscopy and solubility, percentages by dissolution — naming any component the scheme could not resolve and the further test that would resolve it.

3. Worked calculation

The unknown fabric burned like cellulose in part and left hard beads in part, and microscopy showed cotton ribbons together with smooth uniform rods: a cotton/polyester blend. The sulfuric-acid method of ISO 1833 was used, in which the cellulose (cotton) component is dissolved and the polyester remains as residue.

QUANTITY	VALUE
Initial dry mass of specimen	2.00 g
Dry mass of residue after dissolution, washing and drying	0.82 g
Mass dissolved (cotton)	2.00 − 0.82 = 1.18 g

$$\text{Polyester (\%)} = 0.82 / 2.00 \times 100 = 41.0\%$$

$$\text{Cotton (\%)} = 1.18 / 2.00 \times 100 = 59.0\% \quad \text{(check: } 41.0 + 59.0 = 100.0 \text{ ✓)}$$

The sample is therefore reported as **59% cotton / 41% polyester** — a poly-cotton close to the common 60/40 nominal blend, the small departure being within the range expected from manufacturing variation and the uncorrected teaching method. In accredited work, standard correction factors are applied and the result is reported with the specific ISO 1833 part and reagent used; neither refinement alters the identification itself.

> **RESULT**
>
> Qualitative: cotton (twisted ribbon, kidney section, paper-burning, soft ash) blended with polyester (smooth rod, round section, melting to a hard uncrushable bead). Quantitative by selective dissolution: cotton 59.0%, polyester 41.0% by dry mass.

4. Precautions

- Record burning observations before looking at the reference table, and test more than one tuft. A conclusion written from expected answers hides the masked behaviour identification exists to catch.

- Never rely on burning alone for a blend or for the cotton/viscose pair — the test gives a family. Carry every identification through microscopy and, where needed, solubility.
- Observe flame safety: small tufts only, forceps always, no loose sleeves, odour assessed by wafting after removal from the flame. Melted thermoplastic drips cause burns — let them fall clear.
- Handle concentrated sulfuric acid and hot alkali only in the fume cupboard with eye protection and gloves, exactly as the safety sheet directs; never return used reagent to a stock bottle.
- For quantitative work, dry to constant mass at both weighings and wash the residue until neutral. Retained water reports as fibre; retained acid attacks the residue while it dries and reports as loss.
- Remove finishes from the quantitative specimen by the specified preliminary treatment. A resin weighed with the specimen is distributed between the two "components" and falsifies both percentages.

Practice (viva preparation)

1. An unknown fibre curls from the flame, burns slowly with a strong burnt-hair smell, self-extinguishes and leaves a crushable black mass. Name the fibre family and the most likely fibre, and give one test that supports it.
2. Why can burning not distinguish cotton from viscose, and which observation separates them?
3. State the action of (a) acetone, (b) boiling 5% sodium hydroxide, (c) concentrated sulfuric acid, naming the fibre each is used to remove or identify.
4. A 2.00 g cotton/polyester specimen leaves 0.82 g of dry residue after the cellulose is dissolved. Find both blend percentages.
5. Why is quantitative blend analysis carried out on dry mass, and what error follows if the residue is weighed damp?
6. What are correction factors in ISO 1833 work, and why does a teaching result state whether they were applied?
7. A blend burns mainly like cotton. Why does that not prove it is pure cotton?

Answer key

1. A protein fibre — the hair odour, self-extinction and crushable charred mass are the protein signatures — and most likely wool, with silk the alternative. Microscopy supports it: wool shows overlapping surface scales longitudinally and a round-to-oval cross-section, while silk is a smooth rod of triangular section.
2. Both are cellulose, so their flame chemistry is the same — ready ignition, continued burning, paper smell, soft ash. Microscopy separates them: cotton is a twisted flat ribbon with a kidney-shaped section and lumen, whereas viscose shows longitudinal striations and a serrated, lobed cross-section.

3. (a) Acetone dissolves cellulose acetate at room temperature — used to remove/identify acetate. (b) Boiling 5% NaOH dissolves wool (hot dilute alkali destroys protein fibres) — used to separate wool from alkali-resistant cellulose fibres; silk, also protein, is likewise attacked, so the test groups proteins. (c) Concentrated sulfuric acid dissolves cellulose (and attacks polyamide) — used, under standardised conditions, to dissolve the cotton from a cotton/polyester blend, leaving polyester as the weighed residue.
4. Residue (polyester) = 0.82 / 2.00 × 100 = 41.0%. Dissolved (cotton) = (2.00 − 0.82) / 2.00 × 100 = 1.18 / 2.00 × 100 = 59.0%. The percentages sum to 100.0%.
5. Fibres hold very different amounts of moisture, and mass at laboratory humidity includes that water. Percentages must describe fibre, not fibre-plus-weather, so both weighings are made at constant dry mass. A damp residue weighs heavy: the residue component is over-reported and the dissolved component under-reported by the same mass of water.
6. No reagent is perfectly selective: the component intended to survive loses a small, reproducible fraction of its mass during treatment. ISO 1833 methods publish correction factors from collaborative trials so laboratories can restore that loss. Stating whether they were applied tells the reader how the figures were obtained and why a teaching result may sit a fraction away from an accredited one.
7. Burning is dominated by the major component and by any finish: 40% polyester in a cotton blend still presents mainly cellulose flame behaviour, with only the hard beads among the ash hinting at the second fibre, and a flame-retardant finish can suppress burning altogether. Proof of purity requires the full scheme — one morphology only under the microscope, one solubility behaviour, and a dissolution attempt leaving either no residue or no loss.

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